Let’s go back to our touchstone, the problem of a single spin. Any observable of a spin is represented by a 2 × 2 Hermitian matrix, and any such matrix has the form

$$\begin{pmatrix} r & w \\ w^* & r' \end{pmatrix},$$

with the diagonal elements being real and the other two being complex conjugates. The implication is that it takes exactly four real parameters to specify this observable. In fact, there is a neat way to write any spin observable in terms of the Pauli matrices, σ_x, σ_y, and σ_z, and one more matrix: the unit matrix I. As you recall,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$